Dual-Sparse Compressed Systolic Acceleration for SNN workloads

Spiking neural networks (SNNs) are attractive for edge inference because computation is event driven and the core operation is an accumulate (AC) rather than a multiply accumulate (MAC). However, this advantage is eroded by *dual redundancy*:

Temporal redundancy. For a network run over T timesteps, SpinalFlow processes timesteps one after another, so latency grows with T even though each step is cheap. This serialization is a scheduling choice, not a computational necessity: accumulating presynaptic input has no order dependence across timesteps, and only the LIF update does. PTB relaxes the serialization by batching a window of timesteps, but only partially.
SATO and LoAS parallelize across all timesteps, but each leaves a serial cost behind. SATO's log-T adder-search tree locates the firing time by assuming a single spike per neuron, which ties it to temporally coded SNNs. LoAS parallelizes only the timestep loop: within each processing element, its inner-join still consumes matched spike weight pairs **one at a cycle** along the reduction dimension, and the prefix-sum circuits that find those matches account for roughly three-quarters of PE area and over 60% of PE power in its own breakdown.

Spatial redundancy. SNN layers are sparse in spikes and weights. SpinalFlow, PTB and SATO exploit spike sparsity only, so every pruned weight still occupies a PE cycle. LoAS is the first to exploit both, but it must pair nonzero spikes with nonzero weights *at* ***runtime*** through an inner-join circuit. That hardware costs area and power, and it makes execution time depend on where the nonzeros happen to fall.

The gap. Prior designs each close part of the picture. To our knowledge, none combines full temporal parallelism, dual sparsity, and a regular array with no runtime matching.

Our approach. We remove the matching problem before the data reaches the array. Weights and activations are packed by a pre processing stage that runs ahead of the array, off its critical path. Rows and columns with no overlapping nonzero positions are merged into dense tiles, each entry carrying a small index tag. A PE needs only a tag comparison and a small multi-slot accumulator, with T gated accumulators so all timesteps complete in one pass. Packing is lossless (nothing is pruned or retrained), and completion time is a closed form function of the packed dimensions, independent of T and of the sparsity pattern.